

## Ideation Phase Brainstorm & Idea Prioritization

|                      |                                                                              |
|----------------------|------------------------------------------------------------------------------|
| <b>Date</b>          | 04 July 2026                                                                 |
| <b>Team ID</b>       |                                                                              |
| <b>Project Name</b>  | Plugging into the Future: An Exploration of Electricity Consumption Patterns |
| <b>Maximum Marks</b> | 4 Marks                                                                      |

### Step-1: Team Gathering, Collaboration and Select the Problem Statement

#### PROBLEM

**How might we help grid planners, policymakers, and consumers understand and anticipate state-wise and regional power consumption patterns, so that supply, cost, and infrastructure decisions can be made proactively rather than reactively?**

#### Key rules of brainstorming

- Stay on topic
- Defer judgment
- Go for volume
- Encourage wild ideas
- Listen to others
- If possible, be visual

## Step-2: Brainstorm, Idea Listing and Grouping

*Brainstorm — ideas addressing the problem statement, organized by theme:*

| Predictive Analytics                                     | Visualization & Dashboards                              | Grid & Load Management                            | Policy & Renewable Planning                               | Data Quality & Aggregation                                     | Consumer & Industry Tools                        |
|----------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------|
| Build an ML model to forecast next-day state-wise demand | Real-time regional consumption dashboard                | Peak-load early-warning alerts for grid operators | State-wise renewable investment prioritization report     | Automated pipeline to clean and aggregate the raw CSV          | Cost anomaly detection for industrial consumers  |
| Seasonal decomposition of usage trends by state          | Interactive geographic heatmap using latitude/longitude | Region-to-region load balancing simulator         | Growth-rate ranking of states for infrastructure planning | Central data warehouse combining state, region and date fields | Consumer-facing usage insights mobile app        |
| Anomaly detection for unusual consumption spikes         | Year-over-year trend comparison charts                  | Automated load-shedding recommendation engine     | Carbon-offset impact estimator tied to consumption data   | Validation checks for missing or duplicate entries             | Chatbot for querying historical consumption data |

*Group Ideas — clustering related sticky notes into themes:*

| Grouped Theme                         | Ideas Merged Into This Theme                                     |
|---------------------------------------|------------------------------------------------------------------|
| <b>Forecasting &amp; Analytics</b>    | ML demand forecasting, seasonal decomposition, anomaly detection |
| <b>Dashboards &amp; Visualization</b> | Real-time dashboard, geographic heatmap, YoY trend charts        |
| <b>Grid Operations</b>                | Peak-load alerts, load balancing simulator, load-shedding engine |
| <b>Policy &amp; Planning</b>          | Renewable investment report, growth ranking, carbon estimator    |
| <b>Data Infrastructure</b>            | Cleaning pipeline, central data warehouse, validation checks     |
| <b>End-user Tools</b>                 | Cost anomaly detection, consumer app, chatbot                    |

### Step-3: Idea Prioritization

The team placed each idea on the grid below to determine what is important and feasible, in order to decide what to build first.

